COHERENT STATES (aka Glauber states)

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Harmonic oscillator eigenstates $|n\rangle$ have

$$\Delta x \cdot \Delta p = \left(n+\tfrac{1}{2}\right)\hbar$$

$\Rightarrow$ $\underline{n=0}$ state (the G.S.) minimizes the uncertainty relation.

Note that $a|0\rangle = 0$.

Consider now the <u>coherent</u> states

$$\boxed{a|\alpha\rangle = \alpha|\alpha\rangle}$$

α = complex number

$|\alpha\rangle$ = eigenstate of lowering operator.

(a) $$\langle x\rangle = \langle\alpha|x|\alpha\rangle = \sqrt{\frac{\hbar}{2m\omega}}\,\langle\alpha|\,\hat{a}+\hat{a}^+|\alpha\rangle =$$

$$a|\alpha\rangle = \alpha|\alpha\rangle \Rightarrow \langle\alpha|a^+ = \langle\alpha|\alpha^*$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\,(\alpha+\alpha^*)\underbrace{\langle\alpha|\alpha\rangle}_{\sim 1} = 2\sqrt{\frac{\hbar}{2m\omega}}\,\mathrm{Re}(\alpha).$$

$$\langle x^2\rangle = \frac{\hbar}{2m\omega}\,\langle\alpha|\,aa + a^+a^+ + \underbrace{a^+a + aa^+}_{2a^+a+1}|\alpha\rangle$$

$$= \frac{\hbar}{2m\omega}\Big\{\langle\alpha|\alpha^2|\alpha\rangle + \langle\alpha|\alpha^{*2}|\alpha\rangle +$$

$$+ 2\langle\alpha|\alpha^*\alpha|\alpha\rangle + \langle\alpha|\alpha\rangle\Big\} = \frac{\hbar}{2m\omega}\left(\alpha^2+\alpha^{*2}+2\alpha^*\alpha+1\right)$$

$$\Rightarrow \Delta x^2 = \langle x^2\rangle - \langle x\rangle^2 = \frac{\hbar}{2m\omega}\left(\alpha^2+\alpha^{*2}+2\alpha^*\alpha+1\right) -$$

$$-\frac{\hbar}{2m\omega}(\alpha+\alpha^*)^2 = \frac{\hbar}{2m\omega}\cdot 1$$

$$\hat{p} = -i\sqrt{\frac{m\omega\hbar}{2}}\,(a-a^+)$$

$$\Rightarrow \langle\hat{p}_x\rangle = -i\sqrt{\frac{m\omega\hbar}{2}}\,\underbrace{(\alpha-\alpha^*)}_{2i\,\mathrm{Im}\,\alpha} = 2\sqrt{\frac{m\omega\hbar}{2}}\,\mathrm{Im}(\alpha)$$

$[a, a^+] = 1 = a a^+ - a^+ a$

$$\langle \hat{p}^2 \rangle = -\frac{m\omega\hbar}{2} \langle \alpha | \, a a + a^+ a^+ - \underbrace{a a^+ - a^+ a}_{-2a^+a - 1} | \alpha \rangle$$

$$= + \frac{m\omega\hbar}{2} \left(2\alpha^* \alpha + 1 - \alpha^2 - \alpha^{*2} \right)$$

$$\Rightarrow \Delta p^2 = \frac{m\omega\hbar}{2} \left\{ 2\alpha^*\alpha + 1 - \alpha^2 - \alpha^{*2} - \left(-i(\alpha - \alpha^*)\right)^2 \right\} = \frac{m\omega\hbar}{2}$$

but $\left\{ 2\alpha^*\alpha + 1 - \alpha^2 - \alpha^{*2} + \alpha^2 + \alpha^{*2} - 2\alpha^*\alpha \right\} = 1.$

(b) $$\Rightarrow \quad \Delta x \cdot \Delta p = \sqrt{\frac{\hbar}{2m\omega}} \cdot \sqrt{\frac{m\omega\hbar}{2}} = \frac{\hbar}{2} \; !$$

The coherent state is the minimal uncertainty state for any $\underline{\alpha}$.

(c) $$|\alpha\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$$

what are expansion coefficients?

$$\langle m | \alpha \rangle = \sum_{n=0}^{\infty} c_n \langle m | n \rangle = c_m$$

But $|m\rangle = \frac{(a^+)^m}{\sqrt{m!}} |0\rangle \Rightarrow \langle m | = \langle 0 | \frac{(\hat{a})^m}{\sqrt{m!}}$

Thus $c_m = \langle m | \alpha \rangle = \langle 0 | \frac{(\hat{a})^m}{\sqrt{m!}} | \alpha \rangle =$

$$= \langle 0 | \frac{\alpha^m}{\sqrt{m!}} | \alpha \rangle = \frac{\alpha^m}{\sqrt{m!}} \langle 0 | \alpha \rangle = \frac{\alpha^m}{\sqrt{m!}} c_0$$

$$\Rightarrow |\alpha\rangle = c_0 \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

(d) Work out normalization constant c_0:

$$\langle \alpha | \alpha \rangle = 1 = \underbrace{\sum_{m=0}^{\infty} \langle m | \frac{(\alpha^*)^m}{\sqrt{m!}} c_0^*}_{\langle \alpha |} \cdot \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} c_0 |n\rangle =$$

$$= \sum_{m,n} \frac{(\alpha^*)^m \alpha^n}{\sqrt{m!}\sqrt{n!}} c_0^* \cdot c_0 \underbrace{\langle m | n \rangle}_{\delta_{mn}}$$

$$\Rightarrow \langle\alpha|\alpha\rangle = \sum_{n=0}^{\infty} \frac{(\alpha^*)^n \alpha^n}{(\sqrt{n!})^2} |C_0|^2 =$$

$$= |C_0|^2 \sum_{n=0}^{\infty} \frac{(|\alpha|^2)^n}{n!} = |C_0|^2 e^{+|\alpha|^2} = 1$$

e.f. $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$$\Rightarrow |C_0| = \sqrt{e^{-|\alpha|^2}} = e^{-|\alpha|^2/2}$$

as usual, choose the phase $= 0$:

$$C_0 = e^{-|\alpha|^2/2}. \quad \Rightarrow |\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

(e) Time dependence : appears via time-evolution operator $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$

$$|\alpha(t)\rangle = \hat{U}(t)\,|\alpha(0)\rangle$$

But $|\alpha(0)\rangle = \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-|\alpha|^2/2} |n\rangle$

and $\hat{U}(t)\,|n\rangle = e^{-i\hat{H}t/\hbar}|n\rangle = e^{-iE_n t/\hbar}|n\rangle =$

$$= \left(E_n = \left(n+\tfrac{1}{2}\right)\hbar\omega\right) = e^{-i(n+\frac{1}{2})\omega t}|n\rangle$$

Hence $|\alpha(t)\rangle = e^{-i\hat{H}t/\hbar} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-|\alpha|^2/2} |n\rangle =$

$$= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-i(n+\frac{1}{2})\omega t} |n\rangle =$$

$$= e^{-|\alpha|^2/2} e^{-i\omega t/2} \sum_{n=0}^{\infty} \frac{\alpha^n e^{-in\omega t}}{\sqrt{n!}} |n\rangle =$$

$$= e^{-i\omega t/2} e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{(\alpha e^{-i\omega t})^n}{\sqrt{n!}} |n\rangle$$

Observe that $|\alpha(t)\rangle$ looks exactly as $|\alpha(t=0)\rangle$ but with eigenvalue α replaced by $(\alpha e^{-i\omega t})$! [as we know, the global phase $e^{-i\omega t/2}$ does not change the state.]

Since $\Delta x \cdot \Delta p = \frac{\hbar}{2}$ for <u>any</u> eigenvalue α, it follows that the state $|\alpha(t)\rangle$ remains <u>coherent</u> and has minimal possible uncertainty $\Delta x \cdot \Delta p = \frac{\hbar}{2}$ at <u>all times</u>.

(f) Apparently, the ground state of harmonic oscillator $|0\rangle$ is an example of coherent state with eigenvalue $\alpha = 0$.

This is because $a|\alpha\rangle = 0$. $(= 0 \cdot |\alpha\rangle)$.

* These states appear in quantum optics (lasers), Bose-Einstein condensates, path-integral formulation of quantum field theory and many others! Since they <u>minimize</u> the uncertainty relation, we can think of them as closest to the classical description.